Training-Free Group Relative Policy Optimization

- In the traditional GRPO, we sample Goutputs for each input as a group $\{o_1, o_2, \ldots, o_G\}$, obtain the reward for each o, $\{r_1, r_2, \ldots, r_G\}$, and calculate the relative advantage for each group by $\hat{A} = \frac{r_i - \bar{r}}{\sigma_r}$. With the $\hat{A}$, we can use the loss function to update the model. **However, this method is computer-cost.**
- One way to solve this problem is Training-Free Group Relative Policy Optimization. We can maintain an *external experience library* which will be fed along with the query in the prompt to the model. Taking the philosophy of GRPO, we update the experience library while keeping the model frozen.

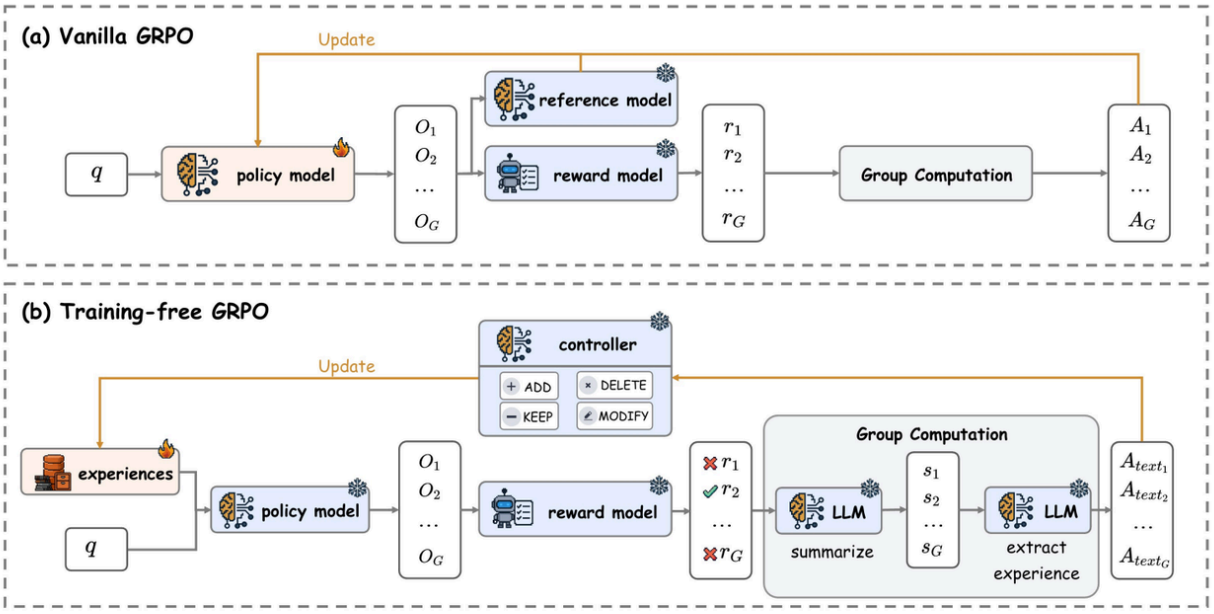

Figure 2. Comparison of vanilla GRPO and Training-Free GRPO.

- Specifically:
  - Like GRPO, we sample $\{o_1, o_2, \ldots, o_G\}$for each input. (*The same as the GRPO.)*
  - **ONLY FOR GROUPS WITH** $\sigma(r) \neq 0$:
    - For each $o_i$in the group, ask the same LLM to generate a summarization with prompt to form $\{s_i\}|_{i=1}^{G}$.
    - For each group, generate group-wise experience by using this prompt. The trajectory will be augmented with s_i and answer (correct/wrong). Along with the group trajectory, problem, (optional) ground truth, and current experiences will also be passed to the LLM.
    - The LLM will then make an *in-group* comparison, select an operation from $[modify,\ add]$, and return a structure called `updates`, like:

    (Corresponds to the advantage $\hat{A}_i$ for $i \in [1, G]$ in GRPO)

    - For the `update` of each group, again, we combine them and pass them into the LLM with this prompt, and the LLM will return the final operation, updated (added) content, and the indexes of to-be-modified old experiences. *(Corresponds to the update of $\pi_\theta$using $\hat{A}$'s.)*
  - Update the experience library based on the results.

- (Graphical demo from the paper)

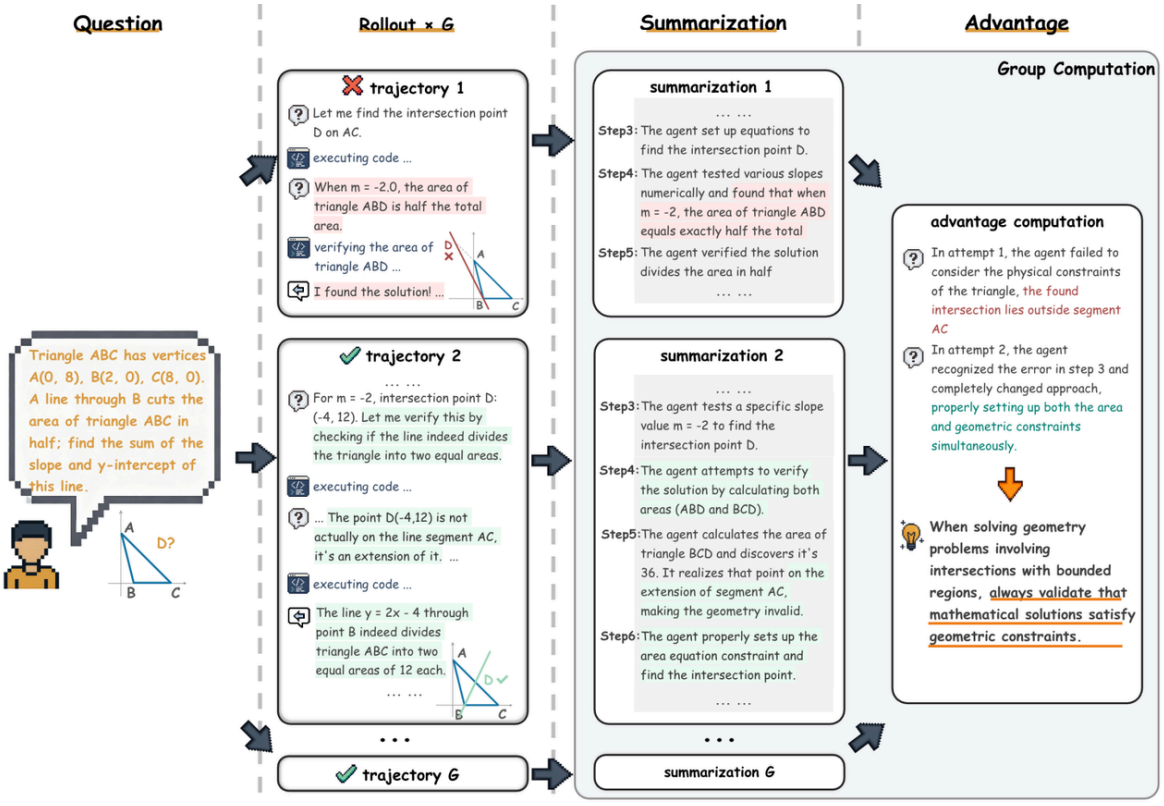

Figure 3. Example of a Training-Free GRPO learning step.

- During the inference time, the original query will be augmented with the experience library by the prompt:
- Advantages
  - **Highly Efficient**: Achieves significant performance gains with minimal training data (e.g., 100 samples) and extremely low cost (~$18), as it requires no parameter updates.
  - **Excellent Generalization**: Avoids overfitting and maintains strong performance across diverse domains by keeping the base model's parameters frozen and simply swapping domain-specific "experiences".
  - **Novel Method**: Shifts optimization from the traditional **parameter space** to the **context space**, using a "semantic advantage" (natural language feedback) instead of a numerical one to guide the model.
  - **Robust**: Remains effective even without ground truth data, as the model can learn by comparing different solution attempts within a group.
- Work Cited and Chat History
  - Paper
  - Repo
  - GPT
  - Gemini

Prompts (with groundtruth):

- Summarization:

- Single Experience
- Experience Update

(Note that in actual implementation, `modify` is updating a single experience entry, and `merge` is updating a group; it can also work as `add` mentioned in the paper.)

Prompts (without groundtruth)